

Project Information:

Project Title: Adaptive Multi-Cell Battery Intelligence Engine

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Branch: Electronics and Communication Engineering (ECE)

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Internship Domain: Embedded Systems

Platform Used: ESP32 (Wokwi Simulator)

Programming Language: C++ (Arduino Framework)

Project Status: Completed Successfully

Mentor Name: Jayanth

Tools Used:

- ESP32
- Wokwi Simulator
- Arduino Framework
- Serial Monitor

Project Objective:

To develop a real-time battery intelligence engine capable of monitoring a simulated 4-cell lithium-ion battery pack, calculating pack parameters, identifying weakest and strongest cells, determining imbalance percentage, and classifying battery health conditions.

Adaptive Multi-Cell Battery Intelligence Engine

1. Introduction

The Adaptive Multi-Cell Battery Intelligence Engine is a real-time battery monitoring system implemented using an ESP32 microcontroller in the Wokwi simulation environment. The system continuously monitors a simulated 4-cell lithium-ion battery pack, calculates important battery parameters, identifies voltage imbalances, and classifies the overall health of the battery pack into different operating states.

The project demonstrates battery monitoring concepts commonly used in Battery Management Systems (BMS) for electric vehicles, energy storage systems, and portable electronic devices.

2. Objective

The objectives of this project are:

- To monitor a simulated 4-cell lithium-ion battery pack in real time.
- To measure individual cell voltages.
- To calculate total pack voltage.
- To determine average cell voltage.
- To identify the weakest and strongest battery cells.
- To calculate imbalance percentage among cells.
- To classify battery health into Healthy, Minor Imbalance, Critical Imbalance, and Pack Failure states.
- To implement a scalable and modular battery monitoring architecture using ESP32.

3. Hardware and Software Requirements

Hardware Components

S.No	Component	Quantity
1	ESP32 Development Board	1
2	Potentiometer	4
3	Connecting wires	12

Software Tool: Wokwi Simulator

4. Circuit Connections

The four potentiometers are used to simulate four lithium-ion battery cells.

Component	Pin	Connected To
Potentiometer 1	Left Pin	GND
Potentiometer 1	Middle Pin	GPIO34
Potentiometer 1	Right Pin	3.3V
Potentiometer 2	Left Pin	GND
Potentiometer 2	Middle Pin	GPIO35
Potentiometer 2	Right Pin	3.3V
Potentiometer 3	Left Pin	GND
Potentiometer 3	Middle Pin	GPIO32
Potentiometer 3	Right Pin	3.3V
Potentiometer 4	Left pin	GND
Potentiometer 4	Middle Pin	GPIO33
Potentiometer 4	Right Pin	3.3V

5. System Architecture

The system continuously reads analog voltage values from four simulated battery cells. The ESP32 converts the ADC readings into voltage values and performs battery analysis. The calculated data is used to identify the weakest cell, strongest cell, imbalance percentage, and battery health condition.

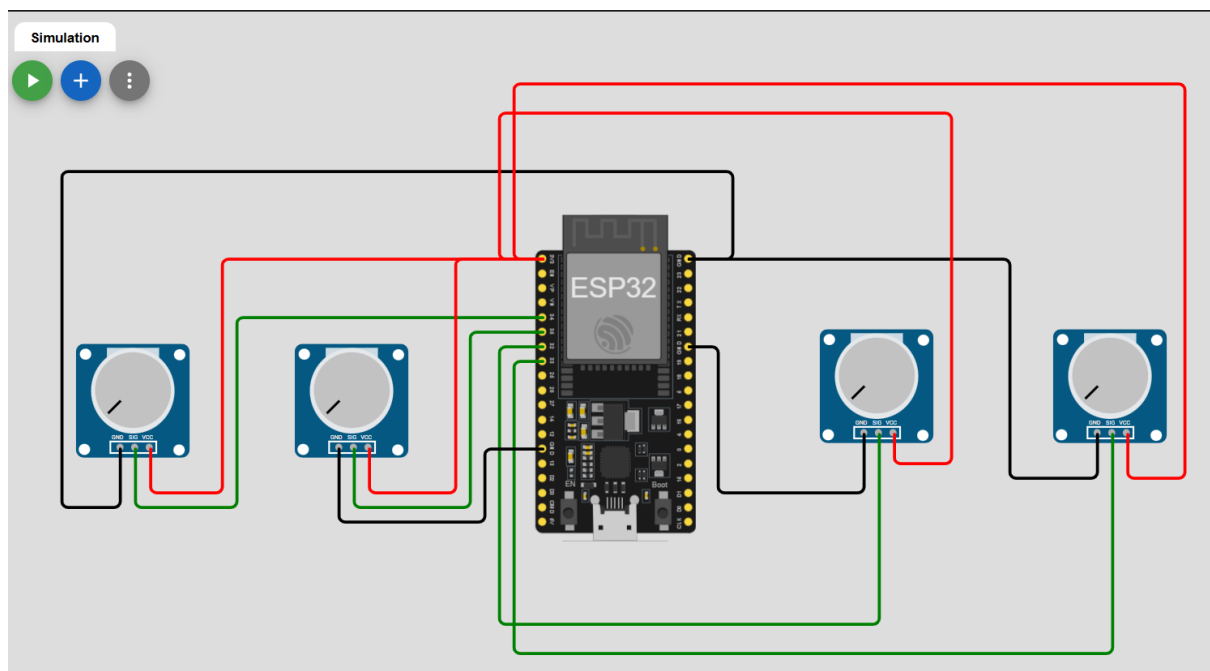


Figure 5.1: Overall System Architecture of Adaptive Multi-Cell Battery Intelligence Engine

5. Working Principle

The Adaptive Multi-Cell Battery Intelligence Engine continuously monitors a simulated 4-cell lithium-ion battery pack using an ESP32 microcontroller. Each lithium-ion cell is represented by a potentiometer in the Wokwi simulation environment.

Step 1: Battery Cell Simulation

Four potentiometers are used to simulate four independent lithium-ion battery cells.

- Potentiometer 1 represents Cell 1
- Potentiometer 2 represents Cell 2
- Potentiometer 3 represents Cell 3
- Potentiometer 4 represents Cell 4

By rotating the potentiometer knob, different battery voltage levels are simulated.

Step 2: Analog Voltage Acquisition

The ESP32 continuously reads analog signals from the following GPIO pins:

Battery Cell	ESP32 GPIO Pin
Cell1	GPIO34
Cell2	GPIO35
Cell3	GPIO32
Cell4	GPIO33

The analog-to-digital converter (ADC) inside the ESP32 converts the incoming analog signals into digital values.

Step 3: Cell Voltage Calculation

The ADC readings obtained from the ESP32 are converted into lithium-ion cell voltages ranging from 3.0V to 4.2V.

The calculated voltages represent the real-time voltage level of each battery cell.

Example:

- Cell 1 = 3.62V
- Cell 2 = 3.58V
- Cell 3 = 3.65V
- Cell 4 = 3.60V

Step 4: Pack Voltage Calculation

The total battery pack voltage is calculated by summing the voltages of all four cells.

Formula:

$$\text{Pack Voltage} = \text{Cell1} + \text{Cell2} + \text{Cell3} + \text{Cell4}$$

Example:

$$\text{Pack Voltage} = 3.62 + 3.58 + 3.65 + 3.60$$

$$\text{Pack Voltage} = 14.45\text{V}$$

Step 5: Average Cell Voltage Calculation

After calculating the total pack voltage, the system determines the average voltage of the battery pack. The average voltage provides an overall indication of the battery condition and serves as a reference for imbalance calculations.

Formula:

$$\text{Average Voltage} = \text{Pack Voltage} \div \text{Number of Cells}$$

Example:

$$\text{Pack Voltage} = 14.45\text{V}$$

$$\text{Number of Cells} = 4$$

$$\text{Average Voltage} = 14.45 \div 4$$

$$\text{Average Voltage} = 3.61\text{V}$$

The calculated average voltage is later used to determine the imbalance percentage between the strongest and weakest cells in the battery pack.

Step 6: Weakest Cell Identification

The system compares all four cell voltages and identifies the cell with the lowest voltage value.

Example:

Cell	Voltage
Cell1	3.62V
Cell2	3.58V
Cell3	3.65V
Cell4	3.60V

Weakest Cell = Cell 2 (3.58V)

The weakest cell is important because it limits the overall battery pack performance.

Step 7: Strongest Cell Identification

The system identifies the cell having the highest voltage value.

Example:

Cell	Voltage
Cell1	3.62V
Cell2	3.58V
Cell3	3.65V
Cell4	3.60V

Strongest Cell = Cell 3 (3.65V)

This information is used to determine voltage imbalance within the battery pack.

Step 8: Imbalance Percentage Calculation

The difference between the strongest and weakest cells is used to calculate the imbalance percentage.

Formula:

Imbalance Percentage = $((\text{Strongest Cell Voltage} - \text{Weakest Cell Voltage}) \div \text{Average Voltage}) \times 100$

The imbalance percentage indicates how uniformly the cells are operating.

Example:

Strongest Cell Voltage = 3.65V

Weakest Cell Voltage = 3.58V

Average Voltage = 3.61V

Imbalance Percentage = $\left(\frac{3.65-3.58}{3.61}\right) * 100 = 1.94\%$

Step 9: Battery Health Classification

Based on the calculated imbalance percentage and voltage thresholds, the battery pack is classified into four operating states.

Healthy State

Conditions:

- All cell voltages are nearly equal.
- Imbalance percentage is less than 2%.

Result:

Battery pack is balanced and operating normally.

Minor Imbalance State

Conditions:

- Small voltage difference exists among cells.
- Imbalance percentage lies between 2% and 5%.

Result:

Battery pack remains operational, but balancing is recommended.

Critical Imbalance State

Conditions:

- Significant voltage difference exists among cells.
- Imbalance percentage exceeds 5%.

Result:

Battery performance may degrade and maintenance is recommended.

Pack Failure State

Conditions:

- One or more cells exceed safe operating limits.
- Cell voltage falls below the minimum threshold or exceeds the maximum threshold.

Result:

Battery pack is considered unsafe for operation.

Step 10: Real-Time Monitoring

The entire monitoring process is repeated continuously in real time.

Any change made to the potentiometer positions immediately affects:

- Cell Voltages
- Pack Voltage
- Average Voltage
- Weakest Cell
- Strongest Cell
- Imbalance Percentage
- Battery Health State

The updated results are displayed on the Serial Monitor every cycle.

6. Results and Observations:

6.1 Healthy State

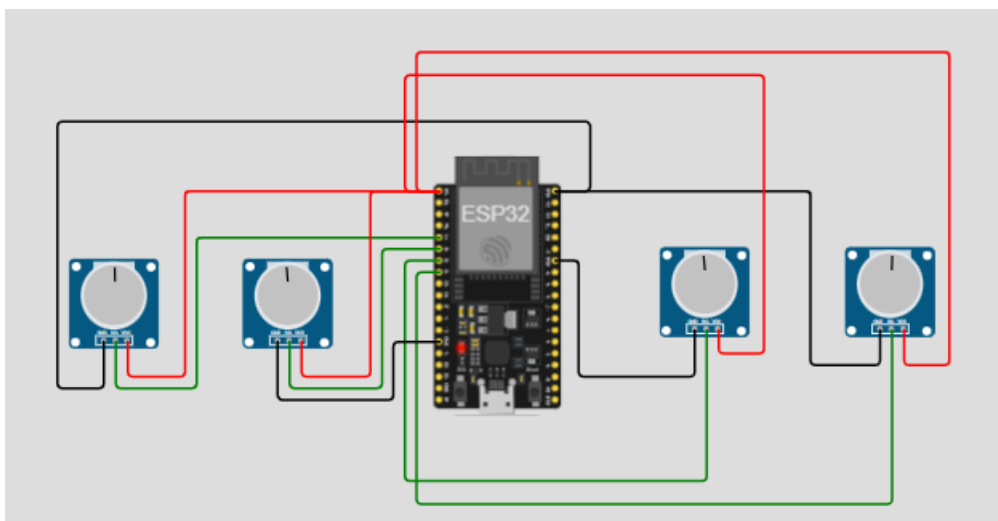


Figure 6.1: Wokwi Circuit Configuration for Healthy State

```
Cell 1 Voltage: 3.60 V
Cell 2 Voltage: 3.58 V
Cell 3 Voltage: 3.58 V
Cell 4 Voltage: 3.61 V
Pack Voltage: 14.36 V
Average Voltage: 3.59 V
Weakest Cell: Cell 2 = 3.58 V
Strongest Cell: Cell 4 = 3.61 V
Imbalance Percentage: 0.86 %
Battery Health State: HEALTHY
```

Figure 6.2: Serial Monitor Output Showing Healthy State

Observation

In this test case, all four potentiometers were adjusted to nearly identical positions, resulting in almost equal cell voltages.

The system calculated:

- Individual cell voltages
- Pack voltage
- Average voltage
- Weakest cell
- Strongest cell
- Imbalance percentage

The calculated imbalance percentage remained below 2%.

The battery health was successfully classified as **HEALTHY**.

Inference

The battery pack is operating under balanced conditions, and no corrective action is required.

6.2 Minor Imbalance State:

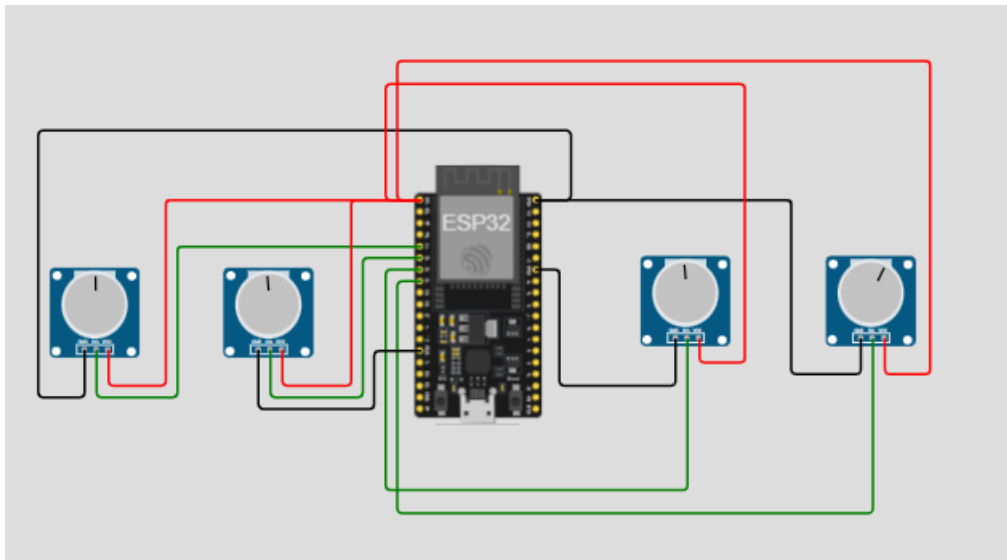


Figure 6.3: Wokwi Circuit Configuration for Minor Imbalance State

```
Cell 1 Voltage: 3.60 V
Cell 2 Voltage: 3.58 V
Cell 3 Voltage: 3.58 V
Cell 4 Voltage: 3.72 V
Pack Voltage: 14.47 V
Average Voltage: 3.62 V
Weakest Cell: Cell 2 = 3.58 V
Strongest Cell: Cell 4 = 3.72 V
Imbalance Percentage: 3.80 %
Battery Health State: MINOR IMBALANCE
```

Figure 6.4: Serial Monitor Output Showing Minor Imbalance State

Observation

In this test case, one potentiometer was adjusted slightly higher than the remaining cells, creating a small voltage variation.

The system successfully:

- Detected the voltage variation
- Calculated pack voltage
- Calculated average voltage
- Identified the weakest cell

- Identified the strongest cell
- Calculated imbalance percentage

The imbalance percentage was found to be between 2% and 5%.

The battery health was correctly classified as **MINOR IMBALANCE**.

Inference

The battery pack remains operational, but cell balancing is recommended to restore uniform cell voltages.

6.3 Critical Imbalance State:

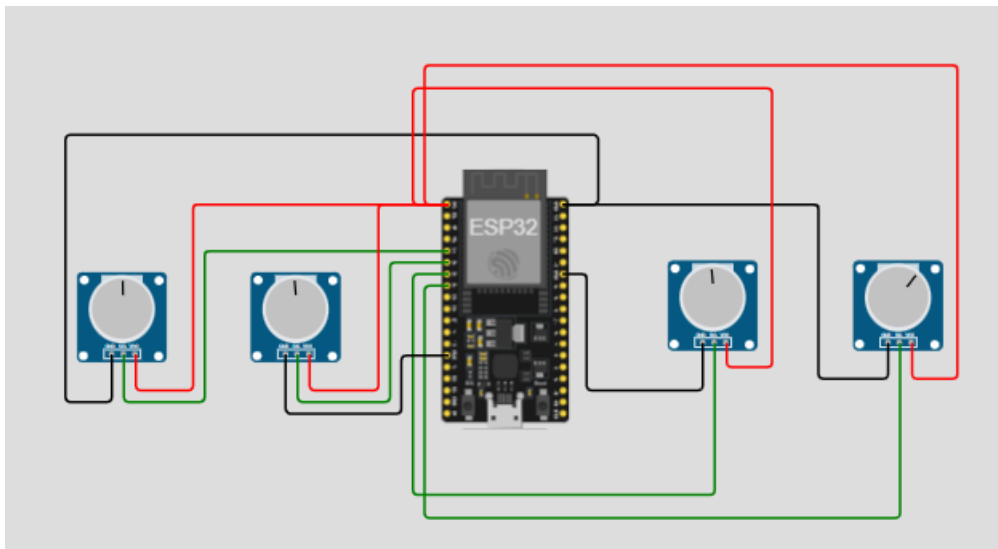


Figure 6.5: Wokwi Circuit Configuration for Critical Imbalance State

```
Cell 1 Voltage: 3.60 V
Cell 2 Voltage: 3.58 V
Cell 3 Voltage: 3.58 V
Cell 4 Voltage: 3.76 V
Pack Voltage: 14.52 V
Average Voltage: 3.63 V
Weakest Cell: Cell 2 = 3.58 V
Strongest Cell: Cell 4 = 3.76 V
Imbalance Percentage: 5.14 %
Battery Health State: CRITICAL IMBALANCE
```

Figure 6.6: Serial Monitor Output Showing Critical Imbalance State

Observation

In this test case, one cell voltage was intentionally adjusted significantly higher or lower than the remaining cells.

The system successfully:

- Detected severe voltage variation
- Identified the weakest cell
- Identified the strongest cell
- Calculated a high imbalance percentage
- Classified the battery condition

The imbalance percentage exceeded 5%.

The battery health was classified as **CRITICAL IMBALANCE**.

Inference

The battery pack is severely unbalanced and continued operation under this condition may reduce battery performance and lifespan.

6.4 Pack Failure State:

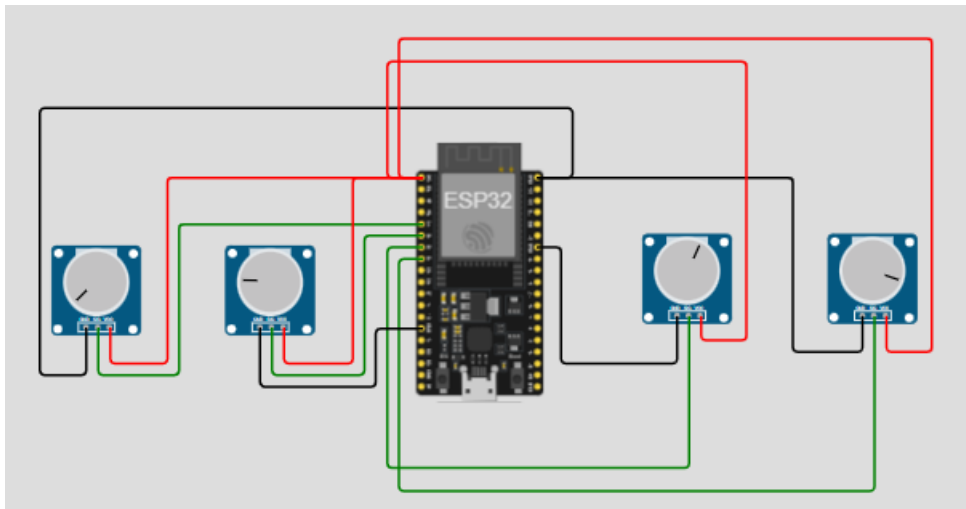


Figure 6.7: Wokwi Circuit Configuration for Pack Failure State

```
Cell 1 Voltage: 3.00 V
Cell 2 Voltage: 3.20 V
Cell 3 Voltage: 3.71 V
Cell 4 Voltage: 4.08 V
Pack Voltage: 13.99 V
Average Voltage: 3.50 V
Weakest Cell: Cell 1 = 3.00 V
Strongest Cell: Cell 4 = 4.08 V
Imbalance Percentage: 30.98 %
Battery Health State: PACK FAILURE
```

Figure 6.8: Serial Monitor Output Showing Pack Failure State

Observation

In this test case, one cell voltage exceeded the predefined safe operating limits.

The system successfully:

- Detected abnormal cell voltage
- Identified the faulty cell
- Calculated battery parameters
- Triggered the Pack Failure condition

The battery health was classified as **PACK FAILURE**.

Inference

The battery pack is operating outside safe limits, and immediate corrective action is required.

6.5 Overall Result

The Adaptive Multi-Cell Battery Intelligence Engine successfully monitored a simulated 4-cell lithium-ion battery pack in real time. The system accurately calculated:

- Individual Cell Voltages
- Pack Voltage
- Average Voltage
- Weakest Cell
- Strongest Cell
- Imbalance Percentage

The system successfully identified and classified all four battery conditions:

1. Healthy
2. Minor Imbalance
3. Critical Imbalance
4. Pack Failure

The obtained results validate the effectiveness of the implemented battery monitoring and battery health assessment algorithm.

8. Performance Parameters Evaluated

The following parameters were successfully monitored:

S.No	Parameter
1	Cell1 Voltage
2	Cell2 Voltage
3	Cell3 Voltage
4	Cell4 Voltage
5	Pack Voltage
6	Average cell Voltage
7	Weakest Cell Detection
8	Strongest Cell Detection
9	Imbalance Percentage
10	Battery Health Classification

9. Conclusion

The Adaptive Multi-Cell Battery Intelligence Engine was successfully designed and implemented using ESP32 in the Wokwi simulation environment. The system continuously monitored four simulated lithium-ion battery cells and calculated important battery parameters such as individual cell voltages, pack voltage, average voltage, imbalance percentage, weakest cell, and strongest cell. Based on predefined thresholds, the battery pack was successfully classified into Healthy, Minor Imbalance, Critical Imbalance, and Pack Failure states. The project successfully demonstrates a real-time battery monitoring and health assessment system suitable for Battery Management System applications.

10. Future Scope

- Integration with LCD/OLED displays.
- Wireless battery monitoring through IoT.
- Automatic cell balancing techniques.
- Cloud-based battery analytics.
- Support for larger battery packs.
- Integration with real Battery Management Systems (BMS).